RC- Series Circuit: An Example

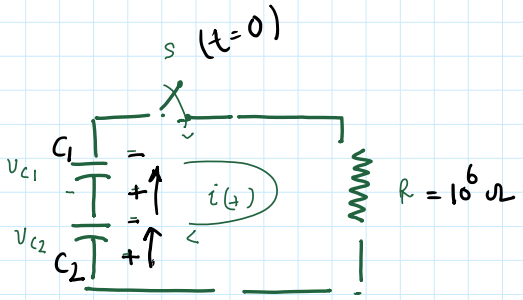
$V_{C10} = -100\,V$

$V_{C20} = 0\,V$

$C_1 = 5\,\mu F$, $C_2 = 20\,\mu F$

$R = 1\,M\Omega$

Determine V_{C1}, V_{C2}, $i(t) = ?$



KVL: $\underbrace{Ri}_{V_R} + \underbrace{\frac{1}{C_1}\int_{-\infty}^{t} i\,dt}_{V_{C10}+V_{C1}} + \underbrace{\frac{1}{C_2}\int_{-\infty}^{t} i\,dt}_{V_{C20}+V_{C2}} = 0$

$$Ri + \frac{1}{C_1}\int_{-\infty}^{0} i\,dt + \frac{1}{C_1}\int_{0}^{t} i\,dt + \frac{1}{C_2}\int_{-\infty}^{0} i\,dt + \frac{1}{C_2}\int_{0}^{t} i\,dt = 0$$

$\int_{-\infty}^{0} i\,dt = V_{C10} = -100$; $\int_{0}^{t} i\,dt = V_{C1}$; $\int_{-\infty}^{0} i\,dt = V_{C20}$; $+ V_{C2}$

$$Ri + (-100) + \frac{1}{C_1}\int_0^t i\,dt + 0 + \frac{1}{C_2}\int_0^t i\,dt = 0$$

$$Ri + \left(\frac{1}{C_1}+\frac{1}{C_2}\right)\int_0^t i\,dt = 100$$

$$Ri + \frac{1}{C_{eq}}\int_0^t i\,dt = 100$$

$$R\frac{di}{dt} + \frac{1}{C_{eq}} i(t) = 0$$

$$\boxed{\frac{di}{dt} + \frac{1}{RC_{eq}} i(t) = 0} \Rightarrow i(t) = K e^{-\frac{t}{RC_{eq}}} + b/a(0)$$

$$i(t) = K e^{-t/RC_{eq}}$$

$e^{-at}\int b e^{at}\,dt$

$C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2} = \frac{5 \times 20}{5+20} = 4\,\mu F$

$R * C_{eq} = 1\times10^6 \times 4\times10^{-6} = 4$

At $t=0$, $V_{C10} = -100$

$$i(t)\Big|_{t=0} = i(0) = K e^{-0} = K$$

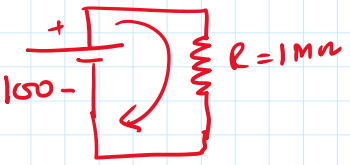
$i(0) = \frac{100}{R} = \frac{100}{1\times10^6}$

$t=0$

$i(0) = \frac{100}{R} = \frac{100}{1\times10^{6}} = 10^{-4}\ A$

$i(t) = 10^{-4}\, e^{-t/RC_{eq}}$

$i(t) = 10^{-4}\, e^{-t/4}$ Amps



$V_{C1} = ? = \frac{1}{C_1}\int_{-\infty}^{t} i\,dt$

$= \frac{1}{C_1}\int_{-\infty}^{0} i\,dt + \frac{1}{C_1}\int_{0}^{t} i\,dt$

$= -100 + \frac{1}{5\times10^{-6}} \int_{0}^{t} 10^{-4} e^{-t/4}\,dt$

$= -100 + \frac{10^{-4}}{5\times10^{-6}} \left.\frac{e^{-t/4}}{(-1/4)}\right|_{0}^{t}$

$= -100 + 20(-4)\left(e^{-t/4} - 1\right)$

$= -100 + 80 - 80\, e^{-t/4}$

$V_{C1}(t) = -20 - 80\, e^{-t/4}$

At $t=0$, $V_{C1} = -100$

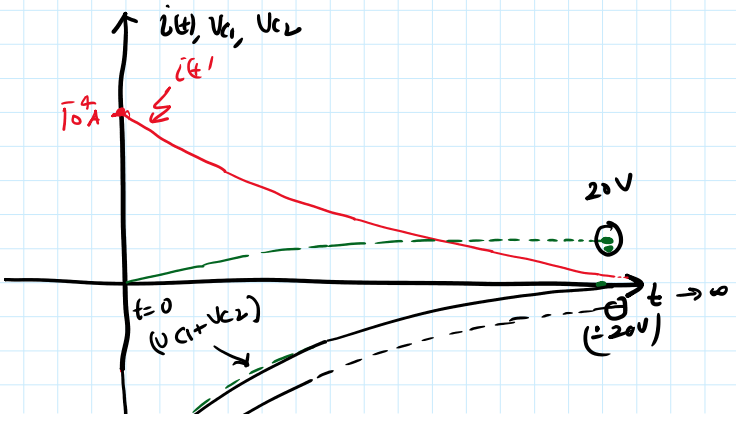
$t=\infty$, $V_{C1} = -20$

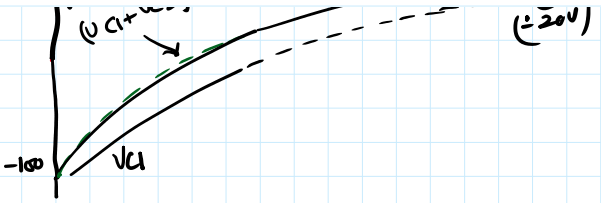
$V_{C2}(t) = \frac{1}{C_2}\int_{-\infty}^{t} i\,dt = \frac{1}{C_2}\int_{0}^{t} i\,dt = \frac{1}{20\times10^{-6}} \cdot \int_{0}^{t} 10^{-4}\, e^{-t/4}\,dt$

$= \frac{10^{-4}\times(-4)}{20\times10^{-6}} \left| e^{-t/4} \right|_{0}^{t}$

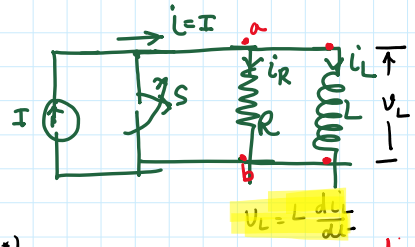
$= -20\left(e^{-t/4} - 1\right)$

$V_{C2}(t) = 20\left(1 - e^{-t/4}\right)$ | $V_{C2}(0) = 0$, $V_{C2}(\infty) = 20$





RL- Parallel Circuit:



KCL:

$I = i_R + i_L$

$= \frac{V_L}{R} + i_L$

$I = \frac{L}{R}\frac{di_L}{dt} + i_L$

or $\frac{di_L}{dt} + \frac{R}{L} i_L = \frac{IR}{L}$ ----(*)

$\frac{di_L}{dt} + a i_L = b$; $a = \frac{R}{L}$, $b = \frac{IR}{L}$

$R i_R = V_L = L\frac{di_L}{dt}$

$i_R = \frac{L}{R}\frac{di_L}{dt}$

$i_L(t) = Ke^{-\frac{R}{L}t} + \frac{IR/K}{R/K}$

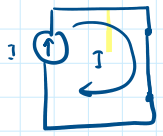
$i_L(t) = Ke^{-\frac{R}{L}t} + I$

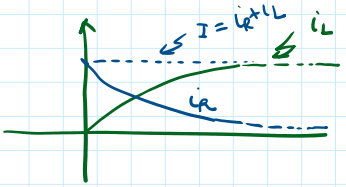
Let $i_L(t)\big|_{t=0} = 0 = K + I$; $K = -I$

$i_L(t) = I\left(1 - e^{-\frac{R}{L}t}\right)$

$i_L(0) = 0$

$i_L(\infty) = I$





$i_R(t) = I - i_L(t)$

$i_R(t) = Ie^{-\frac{R}{L}t}$

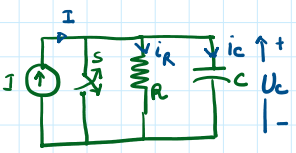
$V_L = V_R = i_R * R = L\frac{di_L}{dt}$

$\boxed{V_L = V_R = IRe^{-\frac{R}{L}t}}$

$V_L(0) = IR$

$V_L(\infty) = 0$

RC Parallel Circuit



$I = i_R + i_c$

$I = \frac{V_c}{R} + C\frac{dV_c}{dt}$

$\frac{dV_c}{dt} + \frac{1}{RC}V_c = \frac{I}{C}$

$\frac{dV_c}{dt} + a V_c = b$ ----- (*)

$i_c = C\frac{dV_c}{dt}$

$i_R = \frac{V_c}{R}$

$V_c(t) = Ke^{-\frac{t}{RC}} + \frac{I/C}{1/(RC)}$

$V_c(t) = Ke^{-t/RC} + IR$

At $t = 0$

$V_c(0) = 0 = K + IR$

$K = -IR$

$\boxed{V_c(t) = IR\left(1 - e^{-t/RC}\right)}$

$V_c(0) = 0$

$V_c(t \to \infty) = IR$

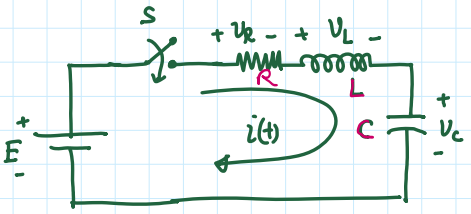
$i_c = C\frac{dV_c}{dt} = C \cdot IR\left(-e^{-t/RC}\right)\left(-\frac{1}{RC}\right)$

$$i_C = C\frac{dv_C}{dt} = C \cdot IR\,(-e^{-t/RC})\left(-\frac{1}{RC}\right)$$

$$\boxed{i_C(t) = Ie^{-t/RC}}$$

RLC Series Circuit

Switch S is closed at $t=0$, $i(0)=0$, $v_C(0)=V_{C0}$



$$v_R + v_L + v_C = E \quad \cdots (1)$$

$$iR + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{0} i\,dt = E$$

$$iR + L\frac{di}{dt} + \underbrace{\frac{1}{C}\int_{-\infty}^{0} i\,dt}_{V_{C0}} + \frac{1}{C}\int_{0}^{t} i\,dt = E$$

$$\Rightarrow iR + L\frac{di}{dt} + \frac{1}{C}\int i\,dt = E - V_{C0}$$

Or $$L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{i}{C} = 0$$

Or $$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0 \quad \cdots (*)$$

A second order differential equation that need to be solved for $i(t)$.

Let $\frac{d}{dt}$ = Differential operator = $S = D$

$$\left(S^2 + \frac{R}{L}S + \frac{1}{LC}\right) I(s) = 0$$

$$\boxed{S^2 + \frac{R}{L}S + \frac{1}{LC} = 0} \Rightarrow \text{Characteristic equation of the ckt}$$

$$S_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

$$S_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$i(t) = K_1 e^{S_1 t} + K_2 e^{S_2 t}$$

Depending upon the value of $\left(\frac{R}{2L}\right)^2$ compared to $\frac{1}{LC}$, the following nature of roots are present.

(A) $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$, two distinct real roots

Ⓐ $\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$, two distinct real roots

Overdamped response

Ⓑ $\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$, two equal real roots

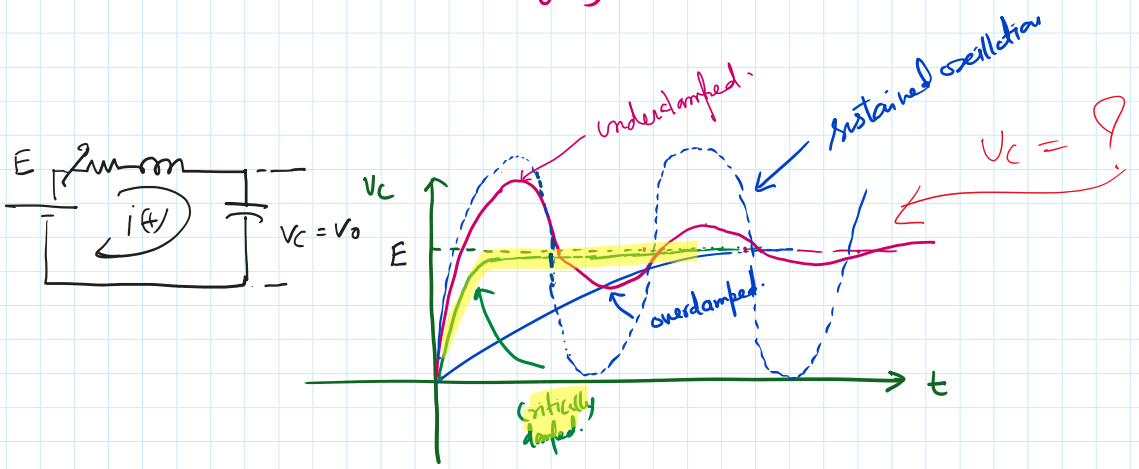
Critically damped

Ⓒ $\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$; complex conjugate roots $\quad s_{12} = -\frac{R}{2} \pm j\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$

Underdamped system

Ⓓ $R = 0 \quad s_{1,2} = \pm j\sqrt{\frac{1}{LC}}$

Roots are imaginary; Sustained oscillation



In the context of this, few important terms are defined.

① Critical resistance, R_c

$$\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$$

$R = R_c$; $R_c = 2L * \sqrt{\frac{1}{LC}} = 2 * \sqrt{\frac{L}{C}}$

$$\boxed{R_c = 2\sqrt{\frac{L}{C}}}$$

② Damping Ratio $\zeta = \frac{R}{R_c} = \frac{R}{2\sqrt{\frac{L}{C}}} = \frac{R}{2}\sqrt{\frac{C}{L}}$

$$\boxed{\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}}$$

③ Natural frequency, $\boxed{\omega_n = \frac{1}{\sqrt{LC}}}$

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$$

$$s^2 + (\ )s + \omega_n^2 = 0$$

$\zeta = \frac{R}{2L} * \sqrt{CL}$

$\zeta = \frac{R}{2L} * \frac{1}{\omega_n}$

$$\boxed{\frac{R}{L} = 2\zeta\omega_n}$$

↓

$s^2 + (\quad)s + \omega_n^2 = 0$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \leftarrow$$

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$= -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - \omega_n^2}$$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$\frac{R}{2L} = \zeta\omega_n$$

$$\frac{R}{2}\sqrt{\frac{C}{L}}$$

$$0 \le \zeta \le 1$$

$$0 < \zeta < 1$$

④ Damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$

$$s_{1,2} = -\zeta\omega_n \pm j\omega_d$$